

INFORMATION SHEET

Order No. 5-00-086
El Dorado Irrigation District
El Dorado Lift Station Equalization Pond

The El Dorado Wastewater Treatment Plant, Sanitation District No. II was replaced by the El Dorado Lift Station and the Mother Lode Force Main to the Deer Creek Wastewater Treatment Facility in the late 1970's. The force main has other smaller connections that pump into the line going towards Deer Creek.

There are two different size pumps at the El Dorado Lift Station. During the 1,900 gpm pumps operation smaller connections down the line are unable to pump into the force main. When the 1,900 gpm pumps cycle off and the 1,000 gpm pumps cycle on, this pump schedule allows the smaller connections to pump into the line while the 1,000 gpm lift station pumps are on. This operational constraint limits the flow that the lift station can pump at one time. Due to I/I problems during storm events, the lift station inflow exceeds the operational limit the station can pump into the force main. Therefore, the excess flow must be diverted to the old Sanitation District No. II North Pond to hold the overflow temporarily until it can be pumped into the force main and on to Deer Creek.

The North Pond is used only as an equalization basin with temporary storage until the I/I from the storm subsides and the wastewater flow resumes at its normal volume.

The North Pond has been recently improved with clearing of excess vegetation, paving, and the reconstructive work listed below:

1. Dewatering the natural spring upstream of the North Pond by the construction of a curtain drain with diversion to south end of the site.
2. Regrading the North Pond slopes to an approximate 2:1 (H:V), and the pond bottom.
3. Scarifying, moisture condition and recompacting the pond sediment to 95% of its maximum dry density at moisture content of 2 to 4% over optimum.
4. Establishment of a vegetative visual screen along the northern limit of the North Pond.

The North Pond's native material has been reworked and compacted to a 95% relative compaction achieving 1.0×10^{-6} cm/sec permeability liner.

Only the North Pond that has been improved is to be used for storage of the overflow. The North Pond occupies a surface area of approximately 1.8 acres. It has a capacity of approximately 3 million gallons and ranges in depth from 7 to 8 feet. With a rain storm that drop five to six inches in a few days the North Pond could fill with as much as about 1.5 million gallons of storm water and wastewater combined.

The surface water drainage is to Slate Creek, which is tributary to Dry Creek, which is tributary to Weber Creek, which is tributary to South Fork of the American River.